MScFE 652 PORTFOLIO MANAGEMENT

Group Work Project # 1

See grading rubric here

Scenario

A portfolio manager is considering the following 10 securities to build a portfolio for one of her clients: **TSLA, WMT, BAC, GS, LLY, MRK, GOOG, META, AAPL, XOM**.

The manager has decided that the correct period to consider in estimating the mean returns and the variance covariance matrix of returns for the given assets is **September 1, 2023 to 30 September, 2025**. She decided to use either **daily** or **weekly data**.

You will also need the data from October 1, 2025 to December 31, 2025 to perform some out-of-sample testing. So plan accordingly when acquiring the dataset.

Part 1 (Standard mean-variance portfolio optimization)

Step 1: Determine the weights of the optimal mean-variance portfolio assuming the following constraints:

a. weights must sum to 1
b. no short-selling is allowed.

Step 2: In a meeting with her client, the portfolio manager is informed that the client does not want any of the assets to represent more than 15% of the portfolio. Determine the new weights of the optimal mean variance portfolio when no asset is allowed to have a weight of more than 0.18.

Step 3: Compare the efficient frontiers in the two instances above. What is the main difference that you notice? (Hint. To answer this question, it might be useful to plot both efficient frontiers on the same graph).

Step 4: Now, use the period October 1, 2025 - December 31, 2025 to provide the performance of both portfolios (Step 1 and Step 2) over that 3 month out-of-sample period. Performance is to be evaluated in terms of **cumulative return**, **Sharpe ratio**, **Max Drawdown** over the out-of-sample period**.**

Important Note: The optimal weights are to be determined using optimization and not simulations. **Using only simulations will be penalized (- 5 points)**

Part 2 (Use of factors to describe the style of the portfolio)

The client is happy with the portfolio found in Part 1, Step 2. Then he asks the portfolio manager to explain to him the exposure of this portfolio to each of Fama-French 5 factors: Market Risk Premium, SMB, HML, CMA, RMW. Essentially, the portfolio manager will estimate the following linear regression model:

$$E[R_p] - R_f = \alpha + \beta_1(Rm - R_f) + \beta_2 SMB + \beta_3 HML + \beta_4 RMW + \beta_5 CMA + \varepsilon$$

where R_f is the risk free rate, R_p is the return on the portfolio and Rm is the return on a suitable market portfolio

Step 4: Explain the meaning/role of each of the 5 factors in the model above. Provide the correlation matrix for the 5 factors and comment on it.

Step 5: Use OLS and a robust model to run the regression and explain specifically how you divided the data into training and testing data.

Step 6: Write a report explaining which factors and how they help explain the exposure of the portfolio returns on the portfolio you selected earlier.

Part 3 (Computing the mean-variance portfolio using simulations)

Now, the portfolio manager decides to investigate whether simulations can replace hard optimization. You will assist her in doing so. More precisely, choose 5 assets out of the 10 given above, with the same time period.

Step 7: Determine the optimal portfolio (for the 5 assets you chose) under the following constraints:

a. Weights have to sum to 1
b. There are no short sales.

You will do so using both optimization and simulation. Discuss how many simulations you need for the optimal weights from simulation to be very close to the ones computed using optimization.

Step 8: Again, use the 5 assets you chose. You will determine the optimal portfolio under the constraints that:

a. No negative weights are allowed,
b. Weights must sum to 1 (i.e. no. leverage is allowed)
c. No assets can be more than 30% of the total portfolio.

As in Step 7, you will do this using both hard core optimization and simulations. How many simulations do you seem to need so that the two sets of weights are close to each other? **Be careful**: the simulation is harder to set up as you have 5 additional box constraints since no assets can have a weight higher than 0.3.

Part 4 (Black-Litterman model)

Step 9: The portfolio manager decides to incorporate his own opinions to build an optimal portfolio. Pretend to be the portfolio manager and implement the Black Litterman model. You are supposed to follow the following steps:

- **(a) Define a suitable market portfolio**. CAREFUL: DO NOT take the 10 assets and declare them your universe and then assume that each has a weight of 1/10 in the basket that is your market portfolio. 10 assets are not a market ! The weights would not be 1/10 anyway. In addition, you need to determine the Sharpe ratio of the market portfolio. There are several ways to proceed (besides the lazy and wrong way I just warned you not to follow). One approach is the **synthetic composite approach**. This method will use your 10 assets and we are going to add an 11th asset which is the ``rest of the S&P 500". The historical returns of this 11th assets are the historical returns of the S&P 500 - the weighted returns of the 10 assets that you have already. The weights of the 10 assets are easy to determine and the weights of the 11th assets is just 1- the sum of those 10 weights. Pay attention to the fact that also your covariance matrix, must use 11 assets and not just 10. If you feel adventurous you can try to come up with a benchmark for your portfolio instead of using the synthetic composite approach I just described, but be prepared to defend your choice.
- **(b) Explain your choice of the parameters τ** . Pay attention that the choice of this parameter is sensitive to the time horizon of the data being used. One

reasonable choice is 1/T. You can use a different one, but you have choices. Just explain them properly.

- **(c) Clearly enunciate your views for the returns** of the assets in your portfolio by writing the matrix P and make sure to explain your choice of their uncertainty in the choice of the matrix Ω.
- **(d) Clearly show your result** for μ_{BL} and Σ_{BL}.
- Now, **determine the optimal weights** for the portfolio using the new parameters μ_{BL} and Σ_{BL}.
- **(e) Finally, discuss the difference between the BL portfolio and the one you obtained in Part 1**. You are allowed and welcome to use portfolios with or without constraints. (For instance, did one or more of your views contribute to changing the weights of some of the assets in a substantial way? (Do you notice if the weights change more or less when there are constraints on the weights of the portfolio?

PAY ATTENTION. Your writeup must be essential but clear and informative. DO NOT limit yourselves to just one page of results without any explanations. **A penalization of 5 points will be assessed for that.**

Groups of 3 students are responsible for all 9 Steps.

Groups of 2 students are responsible for Steps 2, 4, 5, 6, 8, 9

Groups with only 1 student are responsible for Steps 2, 4, 5, 6, 7, 9

Submission requirements and format

READ the WQU Academic Policy on the Use of AI for a proper use of this tool. Instructors heavily penalize any improper use.

One team member submits on behalf of the entire group the following:

1. 1 **zipped folder** including:
    a. One (1) Jupyter notebook (**a single notebook for the entire group and not different notebooks for the different parts**) in .ipynb format The notebook MUST contain also the results and not just the code! No need to show the data you read in, but all the portfolio related snippets of Python code must have the output included.

b. A copy of the same Jupyter Notebook in either .pdf or .html format. Any other supporting materials that you might have used

IMPORTANT: If the submission is not properly done **a penalization of 5 points will be assessed.**

2. **1 PDF document*** from an original paper in Google Docs or LateX.
    a. Use the available Report Template and fill out the required information on the first page, then make sure to include the reports in the PDF along with the template.
    b. Relevant parts of the Jupyter Notebook output such as tables or graphs that you mention in the paper **MUST** be reproduced in the paper itself. No talking about graphs, tables and outputs in general and inviting the reader to look into the Notebook. Your main submission is the report paper and not the Notebook which is like your working canvass.
    c. There is no strict limit on the amount you can write, but please be considerate. There are no extra points for writing a lot. However, there could be a penalty for writing too little. It is usually when students lack clarity about what they are doing that they write a lot, trying to confuse the instructors with a "sea of words". Don't!
    d. Make sure that your paper looks professional: use the same fonts and the same character size throughout the report (use template instructions), center your plots and tables, control that your formulae are properly formatted and absolutely do not copy paste formulae from other sources as they will look different in font and size from the rest of the text.
    e. References are to be listed in the end and whenever a reference is used, it must be indicated in the body of the paper as an in-text citation. Do not provide references as links only. Use the provided In-text Citations and References Guide available in the Student Resource Center.

** **Use Google Docs to collaborate** . Start by uploading the Report Template provided in the Course Overview. Once your report is completed, click File → Download → PDF Document (.pdf) to obtain the copy for your submission.*

*** **Use Google Colab or GitHub to collaborate** in completing the executable Python program.*

*The PDF files must be uploaded **separately** from the zipped folder that includes any other types of files. This allows Turnitin to generate a similarity report.*

Rubric

Your instructor will evaluate your group submission for GWP1 using the following rubric:

Quantitative Analysis (open-ended questions) 60 Points	Technical and Non-technical Reports 40 Points	Writing and Formatting 20 Points
The group is able to apply results, formulas, and their knowledge of theory to real-life finance scenarios by doing the following: ● Providing all the necessary information to support their arguments. ● Presenting arguments that reflect group discussion and research. ● Using authoritative references to support a position and provide updated information ● Concluding with practical takeaways for more insightful financial decision-making	Technical Reports contain 3 parts: 1) summary of key results; 2) interpretation of results; and 3) the recommended course of action that can reasonably follow from those results and interpretations. **Note**: Technical reports will include the technicalities of models, such as names, methods of estimation, parameter values, etc. and exclude generalities about the work done. It should NOT include names of Python code that were used.	A submission that looks professional should include: ● The axes, labels, and scales in graphs. ● No significant grammar errors or typos. ● Organized, well-structured, and easy-to-read document. ● Proper citations and bibliography using MLA format.
	Non-technical Reports contain 3 parts: 1) clear explanation of results; 2) the recommended course of action that follows; and 3) the identification of factors that impact each portfolio. **Note**: AVOID all references to model names, algorithms, and unnecessary details. Instead, focus on the investment decision.	

Revised: April 8, 2026